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CHAPTER 1

INTRODUCTION TO ASPHALT

This chapter introduces the field of asphalt technology. It presents a broad overview of past and present uses of asphalt, as well as an introduction to some of the basic concepts and definitions associated with asphalt.

- **Section 1.1 Historical Review**
  - 1.1.1 Background
  - 1.1.2 Historical Milestones
- **Section 1.2 Asphalt Cement**
- **Section 1.3 Types of Asphalt Pavement Construction**
  - 1.3.1 Plant Mix
  - 1.3.2 Surface Treatments
- **Section 1.4 Design of Asphalt Pavements**
  - 1.4.1 Basic Concepts
  - 1.4.2 Surface Course
  - 1.4.3 Base and Subbase Layers
  - 1.4.4 Determining Required Pavement Thickness
  - 1.4.5 Stage Construction
- **Section 1.5 Overview of the Handbook**

1.1 Historical Review

1.1.1 Background

The word *asphalt* is believed to be derived from the ancient Akkadian term *asphaltic*. The Homeric Greeks adopted the term in the form *asphaltos*, meaning "to make firm or stable." The Akkadian *asphaltic* evolved to the late Latin *asphaltum*, the French *asphalte*, and ultimately the English *asphalt*. From its ancient past to the present, asphalt has been used as cement for bonding, coating, and waterproofing objects. Asphalt truly is one of nature's most versatile products.

Asphalt is man's oldest engineering material, having been used since the dawn of civilization. About 6000 B.C. in Sumeria, there existed a thriving shipbuilding industry that produced and used asphalt. Asphalt was used for mortar in building the ziggurats—the Tower of Babel being one of many ziggurats. A thriving asphalt production existed near the cities of Sodom and Gomorrah. Asphalt was used by the Egyptians as a waterproofing material as early as 2600 B.C. From the Persian word for asphalt, *mumiya*, we derive the word mummy. In the ancient world, asphalt was widely used as mortar for building and paving blocks, as caulking for ships, and as waterproofing in numerous applications.

The early asphalts occurred naturally. They were found in geologic strata as soft, readily workable "mortars" and as hard, friable black veins of rock formations. The soft asphalt material is typified in the Trinidad Lake depository on the island of Trinidad, in Bermudez Lake in Venezuela, and in extensive "tar sands" throughout western Canada. These soft asphalt products, called natural asphalts, were used extensively until the early part of the twentieth century.

In the early 1900s, the discovery of a method of refining asphalt from crude petroleum—as well as increasing demand for better roads because of the overwhelming popularity of the automobile—created an expanding industry. Asphalt appeared to be a cheap and inexhaustible resource for smooth, modern roads and numerous other applications (see Figure 1.1).

As the asphalt paving industry blossomed, the physical properties and character of asphalt needed to be determined. To ensure that asphalt roadways were durable and reliable, numerous tests and procedures were developed in the early 1900s. Probably the first scientific approach to designing asphalt paving mixes was to determine asphalt content based on aggregate voids.

In order for design samples to represent field performance, compacted samples had to contain the desired void and asphalt cement contents. In the mid-1920s, Prévost Hubbard and F. C. Field of the Asphalt Institute developed one of the earliest methods of evaluating the physical properties



Figure 1.1 Pavement Placement in 1907 on Pennsylvania Avenue, Washington, D.C.

of a compacted asphalt mixture. The empirical Hubbard-Field test indicated the stability of a mix by a punching-shear type of failure.

In the 1930s, Francis Hveem, while working for the California Department of Highways, developed a test that ensured that compacted asphalt pavement mixtures were stable and did not bleed excessive asphalt. The principles of the Hveem method are based on triaxial compression and an estimate of asphalt content is determined by the surface area of the aggregate. The Hveem method is still in use.

Also in the 1930s, Bruce Marshall who was an engineer with the Mississippi Highway Department developed an asphalt mix design method that could be used to determine the desired density and asphalt content. With the advent of World War II the U.S. Army Corps of Engineers at the Waterways Experiment Station refined and expanded Mr. Marshall's work for use in the development of the mix design for HMA to use on airfields being exposed to heavy aircraft wheel loads. After World War II the Marshall method continued to be refined for heavier aircraft loads and for use in the design of heavy truck loads. These test procedures continue to be used for the design and quality control of hot mix asphalt (HMA).

In the early 1990s, Superpave was introduced as the result of the research conducted as a part of the Strategic Highway Research Program (SHRP). It has now become the prevailing procedure for the design of HMA mixtures used by the various United States Departments of Transportation.

1.1.2 Historical Milestones

Prehistoric: Intact skeletons of prehistoric animals are preserved to the present day in asphalt surface deposits, La Brea Tar Pits, Los Angeles, California.

3000 B.C.–476: Following discovery of the wheel, the first human-made roads are built in southwestern Asia. Later, the Persians develop roads, in many instances with the use of asphalt. The Romans affect early colonization by means of a military road system that spanned their empire.

1599: Asphalt is first classified, and attempts are made to trace the connection between asphalt and petroleum.

1777: P. C. LeSage conducts the first exposition of the modern theory of the origin of asphalt, in France.

1780: First asphalt composition of prepared roofing appears in Sweden.

1802: Rock asphalt is used in France for floor, bridge, and sidewalk surfacing.

1815: In London, John L. McAdam begins building and improving roads with an interlocked and compacted stone surface. This "macadam" innovation has a profound impact on the roadways of the future.

1824: Asphalt paving blocks are first used.

1829: In Lyons, France, asphalt-mastic pavements are first used.

1837: The first account of asphalt deposits in the United States is revealed in Connecticut. Also, the first exhaustive treatise on the chemistry of asphalt, by J. B. Borissingault in Paris, is published.

1838: Rock asphalt is imported and used in sidewalk construction in Philadelphia.

1844: First composition asphalt roofing is used in the United States.

1858: Near a remote hamlet called Black Creek in southern Ontario, Canada, Charles Tripp and James Miller Williams registered the world's first commercial oil well.

1858: First modern asphalt road, composed of 2 inches (50 millimeters) of compressed asphalt mastic, is laid in Paris, France.

1869: First compacted asphalt pavement is laid in London, England.

1870: First U.S. asphalt roadways are laid in Newark, New Jersey.

1876: First sheet asphalt pavement is laid on Pennsylvania Avenue in Washington, D.C. using imported asphalt from Trinidad Lake.

1899: Barber Asphalt Paving Company is heavily involved in early paving equipment, development, and construction of nearly half of all the early sheet asphalt pavements.

1901: Warren Brothers builds the first successful, reproducible asphalt concrete pavements. They also pioneer the development of early hot mix asphalt facilities.

1902: Approximately 20,000 tons (18,000 tonnes) of asphalt cement is produced in the United States from the refining of petroleum.

1910: First drum mixers and dryer mixers are used at mixing facilities.
1919: The Asphalt Association (later renamed the Asphalt Institute) is founded in New York City to serve the users of petroleum asphalt through programs of engineering, research, and education.
1920: The Bates Experimental Road is built in Illinois.
1921: The Federal Aid Road Act of 1916 is amended, enabling the building and improving of roadways to enable progress in the United States.
1930: The Asphalt Institute is instrumental in the crusade to develop early pavement material and construction procedures.
1941: The U.S. Defense Act of 1941 authorizes the building of roads essential to the war effort.
1946: The post–World War II United States begins a new era of road building. Over the next 10 years, 50,000 miles (80,000 kilometers) of new roads are constructed. During the same period, enough automobiles are manufactured to stretch 200,000 miles (322,000 kilometers) bumper to bumper.
1952: The WASHO Road Test is constructed in Idaho to evaluate the design of flexible pavements.
1956: President Eisenhower signs the historic Federal Aid Highway Act, authorizing the building of 41,000 miles (66,000 kilometers) of the U.S. Interstate and Defense Highways System.
1958–1960: The AASHTO Road Test is built in Ottawa, Illinois.
1960: The full-depth pavement concept is conceived.
1962: First International Conference on the Structural Design of Asphalt Pavements takes place at the University of Michigan in Ann Arbor.
1979: Record-high usage of asphalt in the United States, exceeding 38 million tons (34 million metric tons), is recorded.
1984: Blue Book, *Accelerating the Search for Innovation*, Special Report 202, is published.
1986: Brown Book, *SHRP Research Plans*, is published.
1987: SHRP begins.
1992: TRB SHRP is completed; FHWA begins implementation.
1992: First pilot Superpave projects—two in Wisconsin, one in Indiana, and one in Maryland—are constructed.
1993: AASHTO creates provisional standards.
1994: Expert task groups form.
1995: Regional Superpave centers are established.
1996: First major Superpave projects are constructed.
1996: Lead States Team is formed.
1999: TRB Superpave Committee is formed.
2002: Lead States Team program ends.
2005: TRB Superpave Committee ends.

1.2 Asphalt Cement

Asphalt cement is a dark brown to black cementitious material that consists primarily of bitumens. It is either found in natural asphalt deposits or it is produced by the distillation of petroleum. In the United States, asphalt cement is primarily used for construction and maintenance of highways. The two sources for asphalt cement used in the United States are natural deposits and refined petroleum asphalt.

Natural asphalts range from a relatively soft material to a hard black material found in the veins of a rock formation. The two main sources for the soft material are Trinidad Lake asphalt from Pitch Lake (also known as Trinidad Lake) in Trinidad and Bermudez asphalt from Venezuela. The Bermudez deposit covers 1,100 acres and is estimated to contain about 6 million tons of asphalt. During the last twenty years of the nineteenth century, these natural asphalts were used to construct miles of streets and roads. In the 1880s, Washington, D.C., led the world in the number of miles of asphalt streets in the world.

Of the number of sources of hard asphalt, the most commercially viable today is the Gilsonite deposit in Utah. There are also some sources of rock asphalt, which is rock that has been impregnated with asphalt. The largest deposits of this material are KyRock, which was produced in Kentucky during the 1950s and 1960s, and the White Mines in southwestern Texas. Mining of these rock deposits is similar to that of an aggregate quarry. The processed rock asphalt has been used by many DOTs with a small amount of cutback or emulsified asphalts as a cold mix for maintenance patching.

Petroleum asphalts are produced by refining crude petroleum. The primary process used is distillation. The crude petroleum is heated in a large furnace to about 650°F (345°C) and is distilled (see Figure 1.2). The lighter products (gasoline and naphtha) boil off at the lowest temperatures and the heavy gas oils at the higher temperatures. The heavy residual remaining after this process is completed is a *residuum*, which is further refined into the various grades of asphalt cement.

Three types of asphalt used in asphalt pavement construction are:

- Asphalt cements
- Emulsified asphalts
- Cutback asphalts

As previously discussed, asphalt cements are obtained from the distillation of crude petroleum. At room temperature, they are a semi-solid, very viscous material. At high temperatures, they are liquid. Asphalt cement is primarily used in the production of hot mix asphalt (HMA) for use in road construction. HMA is used as the surfacing material for approximately 96 percent

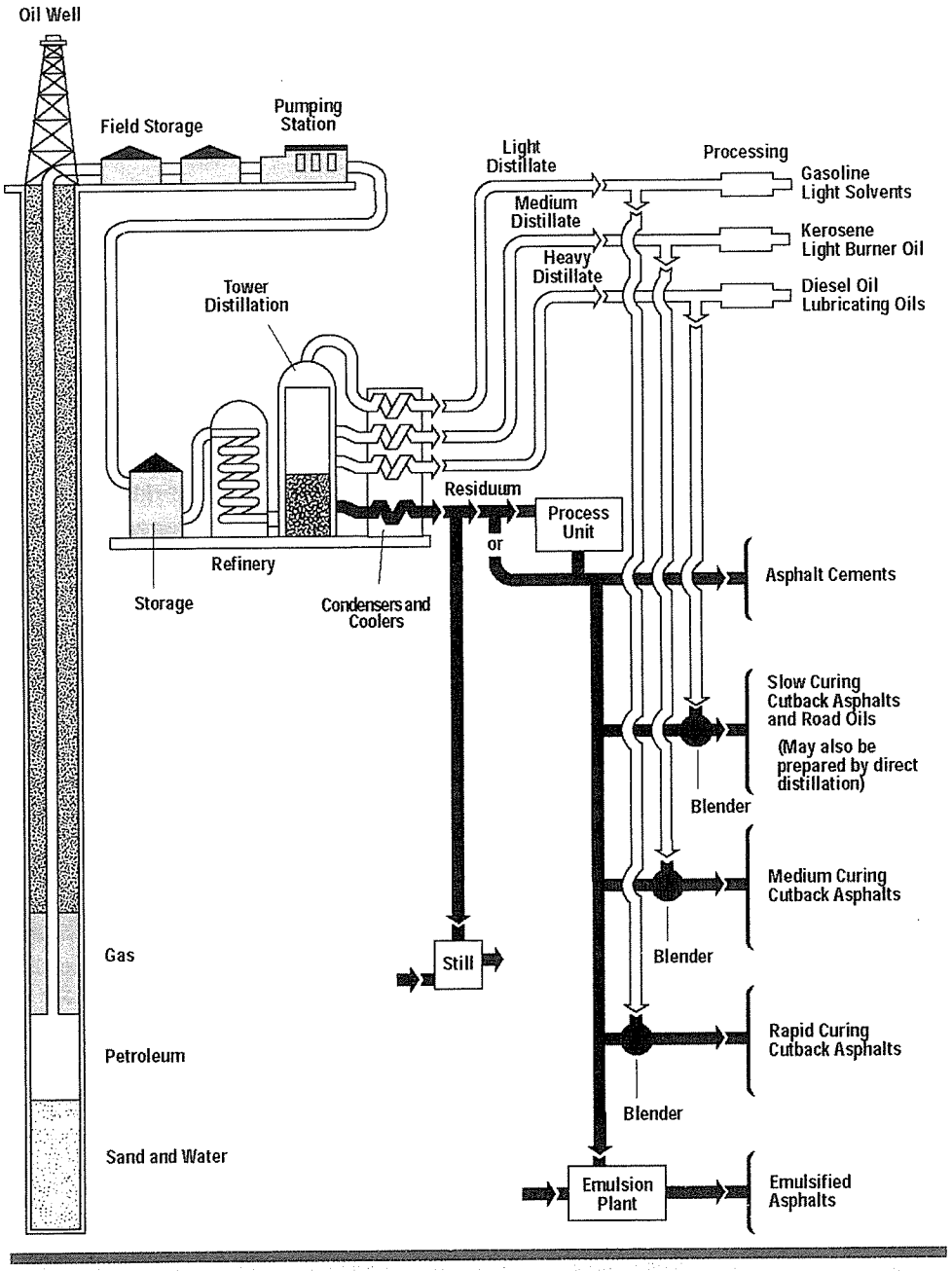


Figure 1.2 Petroleum Asphalt Flow Chart

of the road system in the United States. Several grades of asphalt cement are used, and they are primarily graded according to their consistency. The three grading systems in common use throughout the world for the purchase of asphalt cement are the penetration grading system, the viscosity (both original viscosity and aged residue), and the Superpave system. The grading and testing of asphalt cements is discussed more thoroughly in Chapter 2, "Petroleum Asphalt."

In the last two decades, the use of modified asphalt binders has seen increased use. Modified asphalt binders may be produced in a number of ways including polymer and chemical modification, although polymer modification is the most prevalent.

Emulsified asphalts are a mixture of asphalt cement, water, and an emulsifying agent. Asphalt will not dissolve in water, so hot asphalt cement and water containing the emulsifying agent (such as soap) are passed under high pressure through a colloid mill to produce small droplets of asphalt cement, which are suspended in water. The emulsifying agent imparts an electrical charge on the surface of the droplets, which causes the droplets to repel one another. This prevents the globules from coalescing. Emulsified asphalts are liquid at room temperature. Emulsified asphalts are specified in ASTM D 977 and ASTM D 2397 (AASHTO M 140 and M2). They can be divided into two categories: anionic (electronegatively charged asphalt droplets) and cationic (electro-positively charged asphalt droplets). They are further categorized according to their setting rate—rapid, medium, and slow-setting grades. The testing and properties of asphalt emulsions are further discussed in Chapter 2.

Cutback asphalts are liquid asphalts that are manufactured by mixing asphalt cement with a petroleum solvent, such as gasoline or naphtha (which produces rapid-curing cutbacks), kerosene (which produces medium-curing asphalts), or diesel oil (which produces slow-curing asphalts). They are primarily used for maintenance in the manufacture of patching materials. The testing and properties of cutback asphalts are further discussed in Chapter 2.

1.3 Types of Asphalt Pavement Construction

1.3.1 Plant Mix

1.3

Asphalt paving mixtures or hot mix asphalt is manufactured in a central mixing facility (see Figure 1.3). The workhorse of the asphalt industry is the dense-graded HMA mixture. It consists of a well-graded, high-quality aggregate and asphalt cement. The two components are mechanically mixed while hot to ensure good coating of the aggregate and consistency of the mix. In order to ensure the quality of the resultant mix it is necessary to carefully proportion the blend of several different aggregates to consistently produce a uniform final gradation that meets the requirements of local specifications.

Typically, the asphalt and aggregate are heated to 250°–325°F (121°–163°C). The hot mixture, kept hot during transit, is hauled to the construction site, where it is spread on the roadway by an asphalt-paving machine. The smooth layer placed by the paver is compacted by rollers to proper density before the asphalt cools (see Figure 1.4).

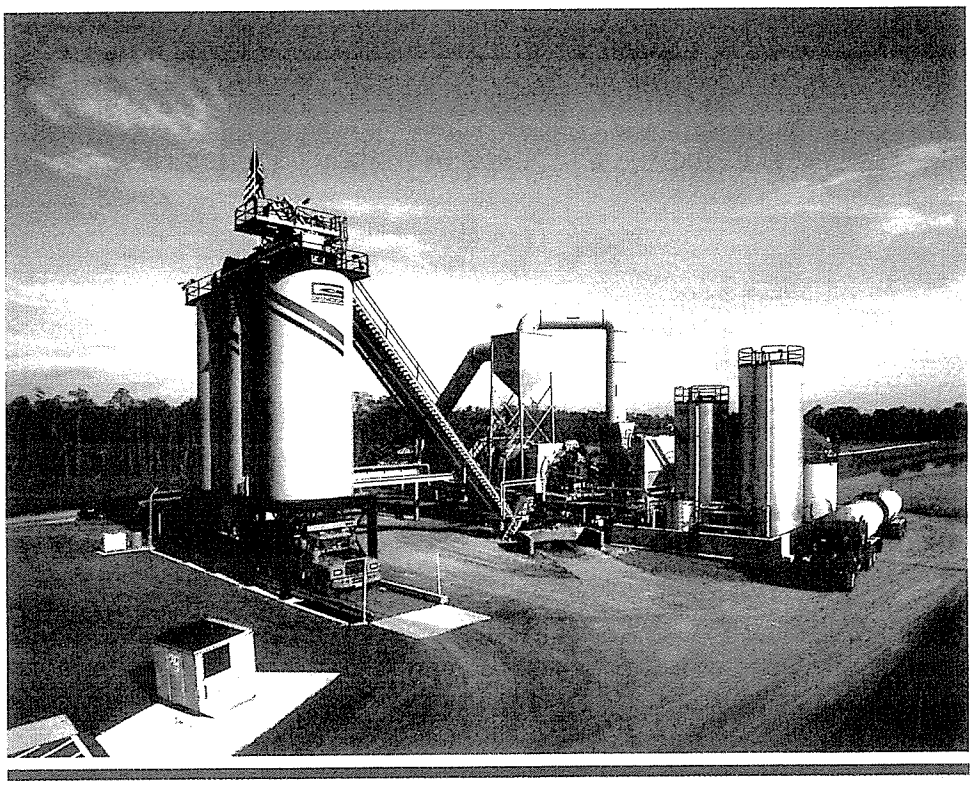


Figure 1.3 Hot Mix Asphalt Manufacturing Facility



Figure 1.4 Placement of Hot Mix Asphalt

In the 1990s, stone matrix asphalt (SMA) was introduced in the United States and the use of open-graded friction courses (OGFC) was expanded.

Stone matrix asphalt is a durable mixture that provides high resistance to permanent deformation. An SMA mixture is generally used on very high-volume roadways or in high-volume intersections. SMA is a gap-graded mixture containing a large proportion of coarse aggregate (about 70 percent) and just enough fine aggregate to fill the voids in the coarse aggregate. The absence of mid-sized aggregate in this type of mix results in stone-to-stone contact, which gives these mixes a high degree of resistance to plastic flow under traffic (rutting). In addition, inorganic filler (typically lime) and fibers (typically cellulose) are added to the mix to make the fine aggregate portion of the mix into a mastic. The mix is also typically made with polymer modified asphalt (PMA). The PMA improves both the strength and the durability of the SMA mix. The inclusion of PMA, fibers and filler is required to prevent drainage of the asphalt cement during transport and placement. This mixture has a surface appearance similar to that of an open-graded friction course; however, it has low in-place air-voids similar to that of a dense-graded HMA. In addition, it has a thicker film of asphalt cement on the coarse aggregate particles, which enhances durability.

An open-graded friction course is another gap-graded mix that is not dense graded but has permeable voids in the range of 16 to 23 percent. These voids allow drainage through the surface of an HMA pavement during heavy rainstorms. The rainwater will drain vertically through the OGFC to an impermeable, underlying layer and then laterally to the day-lighted edge of the OGFC. In addition to minimizing hydroplaning and providing a

high friction resistance on wet pavements, an OGFC layer will also provide a reduction of splash and spray; enhanced visibility of pavement markings; reduced night-time surface glare in wet weather; and reduced tire-pavement noise. The aggregate gradation for an OGFC is generally of one size.

1.3.2 Surface Treatments

Asphalt surface treatments are widely used methods for asphalt pavement construction and maintenance. There are several forms of surface treatments. Properly designed, they are economical, easy to place, and long lasting. They seal and add life to road surfaces, but each type has one or more special purposes. A surface treatment is not a pavement in itself. Rather, it provides both a protective cover that helps to resist traffic abrasion and a waterproof cover over the underlying pavement structure. Surface treatments add little or no load-carrying strength and, therefore, they are not normally taken into account in computing the load-carrying capacity of a pavement. Although a surface treatment can provide an excellent surface if used for the correct purpose, it is not a solution for all pavement problems.

Asphalt surface treatment is a broad term embracing several types of asphalt applications with or without a cover of mineral aggregate to any type road or pavement surface. It is usually less than 1 inch (25 millimeters) thick. An asphalt seal coat, or simply seal coat, is a thin asphalt surface treatment applied to a paved surface. The types of surface treatments include:

- **Spray-applied seal.** This is an application of a diluted asphalt emulsion without a cover aggregate used to seal and enrich the asphalt pavement surface and to seal minor cracking.
- **Chip seal.** This is an application of asphalt followed immediately by an aggregate cover with two layers (referred to as double-chip seal), or three layers, which is called a triple-chip seal.
- **Slurry seal.** This is a mixture of fine aggregate, asphalt emulsion, water, and mineral filler (generally portland cement).
- **Microsurfacing.** This is sometimes referred to as a polymer-modified slurry seal. It differs from a slurry seal, which is cured through a thermal process. With microsurfacing, the curing process is chemically controlled.

1.4 Design of Asphalt Pavements

1.4.1 Basic Concepts

The basic idea in building a road or parking area for all-weather use by vehicles is to prepare a suitable subgrade or foundation, provide necessary drainage, and construct a pavement that will:

- have sufficient total thickness and internal strength to carry expected traffic loads;
- be sufficiently impermeable to prevent the penetration or internal accumulation of moisture; and
- have a top surface that is smooth, skid resistant, and resistant to wear, distortion, and deterioration by weather and deicing chemicals.

The subgrade ultimately carries all traffic loads. Therefore, the structural function of a pavement is to support a wheel load on the pavement surface and transfer and spread that load to the subgrade without exceeding either the strength of the subgrade or the internal strength of the pavement itself (see Figure 1.5).

Figure 1.5 shows wheel load, W, being transmitted to the pavement surface through the tire at an approximately uniform vertical pressure, P_0. The pavement then spreads the wheel load to the subgrade so that the maximum pressure on the subgrade is only P_1. By proper selection

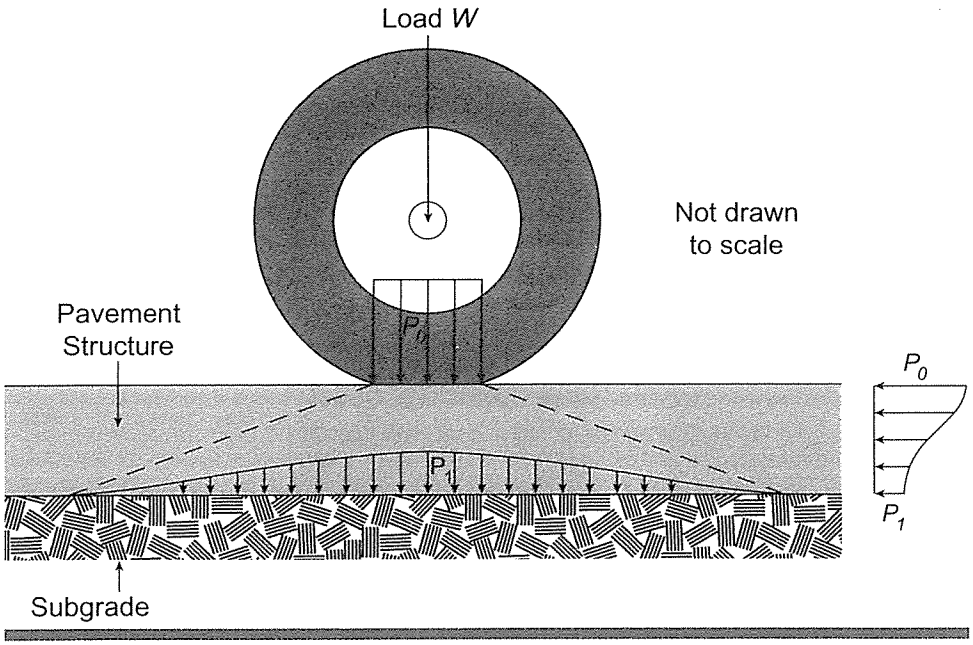


Figure 1.5 Spread of Wheel Load through Pavement Structure

of pavement materials and with adequate pavement thickness, P_1 will be small enough to be easily supported by the subgrade.

1.4.2 Surface Course

Asphalt pavement is a general term applied to any pavement that has a surface constructed with asphalt (see Figure 1.6). Normally, it consists of a surface course (layer) of mineral aggregate coated and cemented with asphalt, and one or more supporting courses, which may be the following:

- Asphalt base, consisting of asphalt-aggregate mixtures
- Crushed stone (rock), slag, or gravel
- Portland cement concrete

An asphalt pavement structure consists of all asphalt-bound layers and unbound (gravel or crushed stone) layers above the prepared subgrade. The upper or top layer consists of an asphalt surface. This surface can be a

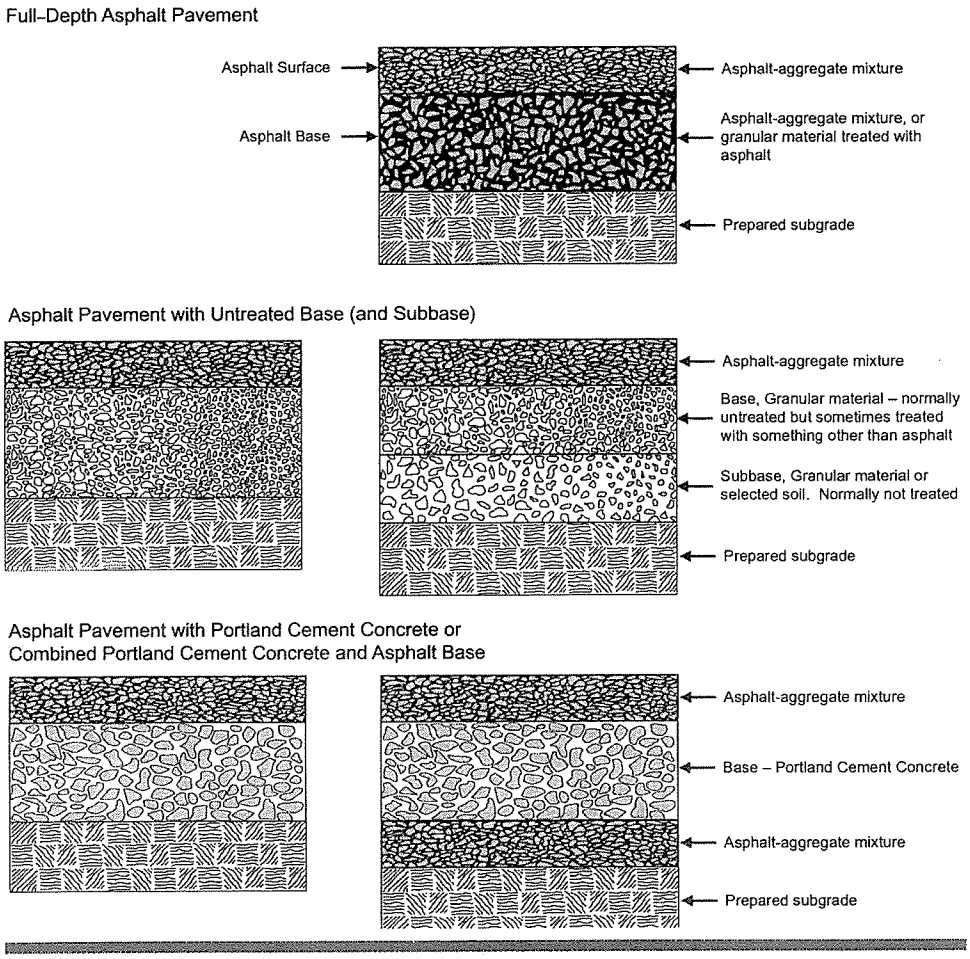


Figure 1.6 Asphalt Pavement Cross-Sections Showing Some Common and Typical Asphalt Pavement Structures

chip seal (for low-volume roads) or a high quality HMA for higher-volume roads and parking lots. The surface of an asphalt pavement exposed to vehicular traffic must be tough to resist distortion and provide a smooth, skid-resistant riding surface. It must be waterproof and sloped to shed surface water to the roadside, thus protecting the entire asphalt pavement structure and the subgrade from the weakening effects of moisture. It must resist wear caused by traffic and still retain necessary anti-skid properties. It must also be bonded to the layer or course beneath it.

1.4.3 Base and Subbase Layers

Although a variety of bases and subbases may be used in asphalt pavement structures, they generally consist of compacted granular materials (such as crushed rock, slag, gravel, sand, or a combination of such) or stabilized soil. One of the main advantages of asphalt pavements is the economy achieved by using locally available materials.

Generally, it is preferable to treat the granular material used in bases. The most common treatment is to mix emulsified, cut-back, or expanded asphalt with the granular material, thus producing an asphalt-bound base.

When the entire pavement structure above the subgrade consists of asphalt mixtures, it is called a full-depth asphalt pavement. This is generally considered the most cost-effective, dependable type of pavement for today's traffic.

The base and subbase are structural elements of the pavement. In conjunction with the overlying asphalt surface, their purpose is to distribute traffic wheel loads over the subgrade or foundation (see Figure 1.5). To perform this function, bases and subbases must be built with necessary internal strength properties. In this respect, full-depth asphalt pavements have a special advantage over pavements with granular bases. Figure 1.7, for example, shows how wheel load, *W,* slightly deflects the pavement structure, causing both tensile and compressive stresses within the pavement. Asphalt pavement layers have both tensile and compressive strength to resist these internal stresses. Untreated granular bases have no tensile strength. Therefore, asphalt bases spread the wheel load over broader areas than do untreated granular bases. As a result, less total pavement structure thickness is required for an asphalt base. This applies to all loading conditions.

1.4.4 Determining Required Pavement Thickness

A significant advance in highway engineering is the realization and demonstration that the structural design of asphalt pavements is similar to the problem of designing any other complex engineering structure. When asphalt pavement was first introduced, determining the proper thickness was a matter

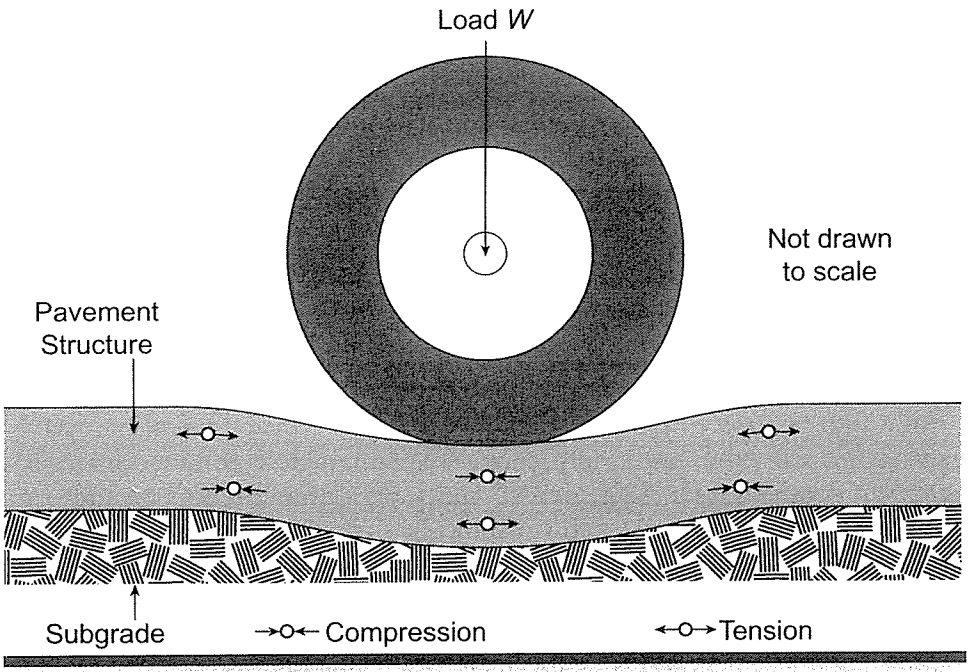


Figure 1.7 Pavement Deflection Results in Tensile and Compressive Stresses in Pavement Structure

of empiricism and opinion based on experience. Similarly, the same situation once prevailed in determining the dimensions of masonry arches and of iron and steel structures. However, these early techniques have long since yielded to engineering analysis. Based on comprehensive analysis of vast volumes of accumulated data, the structural design of asphalt pavement has now evolved into a reliable engineering procedure. Research, aimed at further refinements and a fully mechanistic design procedure, is continuing.

There is no standard thickness for a pavement. Required total thickness is determined by engineering design procedures. Factors considered in the procedures are the following:

- Traffic to be served initially and over the design service life of the pavement
- Strength and other pertinent properties of the prepared subgrade
- Strength and other influencing characteristics of the materials available or chosen for the layers or courses in the total asphalt pavement structure
- Any special factors peculiar to the road being designed, such as environmental factors

1.4.5 Stage Construction

Because weight and traffic volume will normally increase each year, pavement originally built thick enough to handle immediate traffic volumes may not be thick enough and strong enough to handle future needs. With